



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Determine the Whole Given the Part and Percent

Lesson 4-5 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can determine the whole when I know a part and the percent that part represents.

Language Objective: I can explain how I found the whole using the words part, whole, percent, and equation.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Part

Whole

Double number line

Equation

Percent



Tap any word to see its meaning and a picture.

1

How much of the base you get — your answer — is the

2

The full amount a percent is measured against is always

 %

3

Two parallel lines pairing amounts with their percents form a

4

The percent number sentence is part = ×

5

The word that means “per hundred” is .

Write About the Math

What did the car cost when they bought it, and how does your method show it?

¿Cuánto costó el carro cuando lo compraron, y cómo lo muestra tu método?

Say your answer to a partner first. Then write it, using at least two of these words:

☐

Part

Parte

☐

Whole

Total

☐

Double number line

Recta numérica doble

☐

Equation

Ecuación

☐

Percent

Porcentaje

Model: The original price is the whole (100%), and \$42 is only part of it (60%), so the whole must be more than \$42. — Students reason from $60\% < 100\%$ to whole $>$ part, without computing anything.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The \$26,600 is the ____ and the purchase price is the _____. Written as an equation, $0.7v =$ _____. Dividing both sides by _____ gives $v =$ _____. The whole is _____ than the part, which makes sense because 70% is less than 100%.
- My answer is _____ because _____.

Mi respuesta es _____ porque _____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is _____.
- My evidence is _____, so _____.

Mi afirmación es _____.

Mi evidencia es _____, entonces _____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Part
2. **Notes 2:** Whole
3. **Notes 3:** Double number line
4. **Notes 4:** Equation
5. **Notes 5:** Percent
6. **Watch for this mistake:** *A common mistake when finding the whole is multiplying by the percent instead of dividing. For "\$42 is 60% of the original price," computing $0.6 \times 42 = \$25.20$ makes the original price SMALLER than the sale price — impossible for a discount. Since $\text{percent} \times \text{whole} = \text{part}$, the whole is found by dividing: $42 \div 0.6 = \$70$. Whenever the percent is less than 100%, the whole must come out larger than the part, so a smaller answer is the tell.*
7. **Try It 1:** — *"Of what number?" signals a missing whole. "What is ___% of ___?" hands you the whole and asks for the part.*
8. **Try It 2:** — *Every one is the same move: divide the part by the percent written as a decimal.*